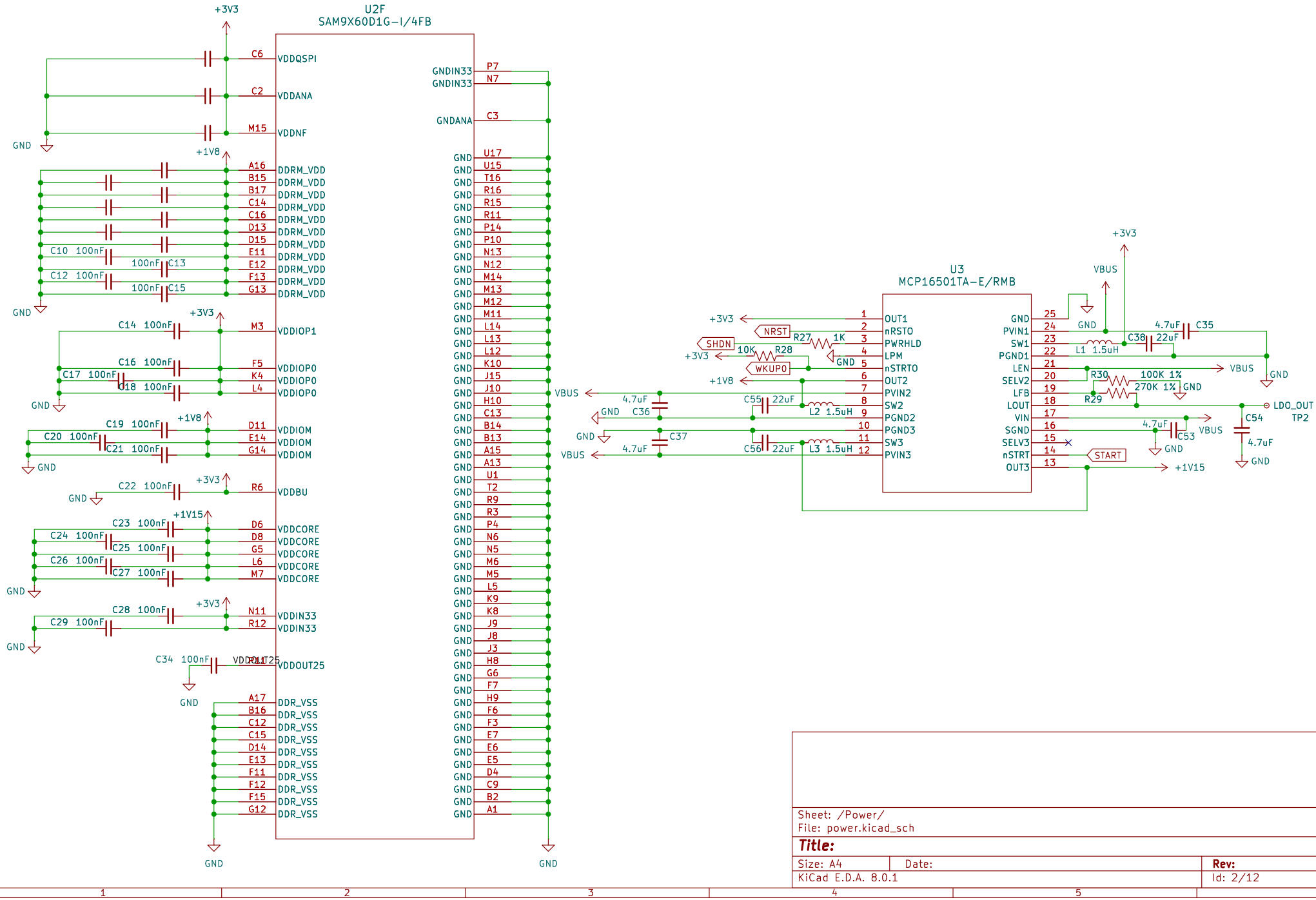
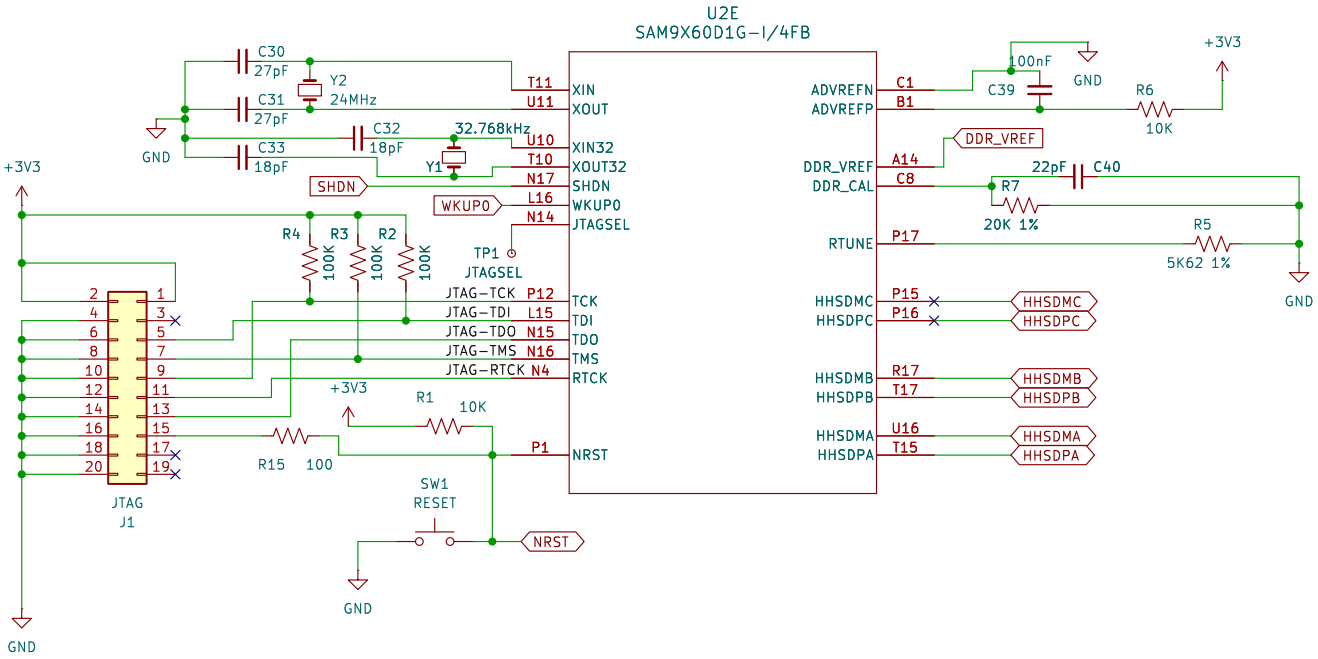
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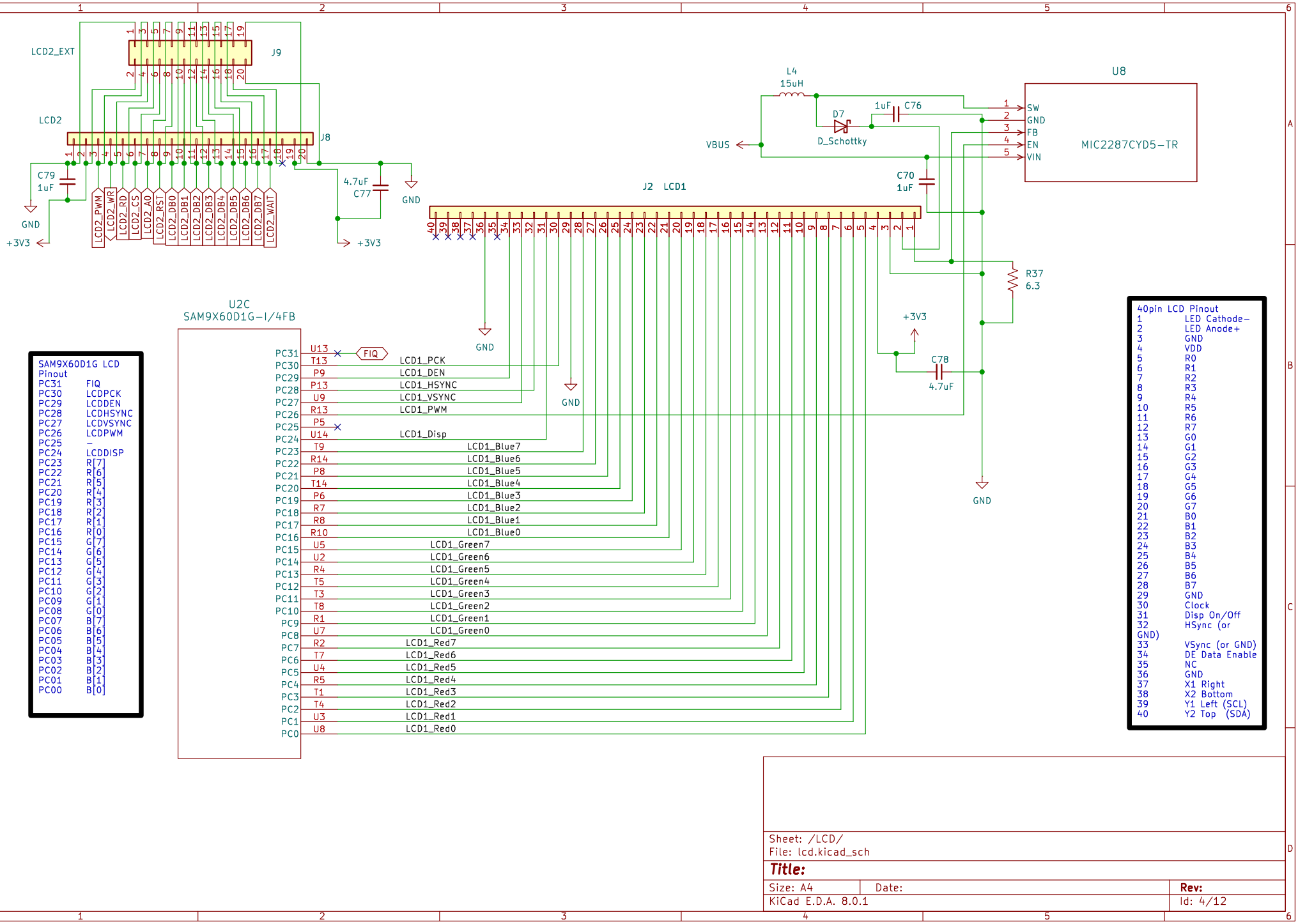
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KiCad E.D.A. 8.0.1		Id: 1/12

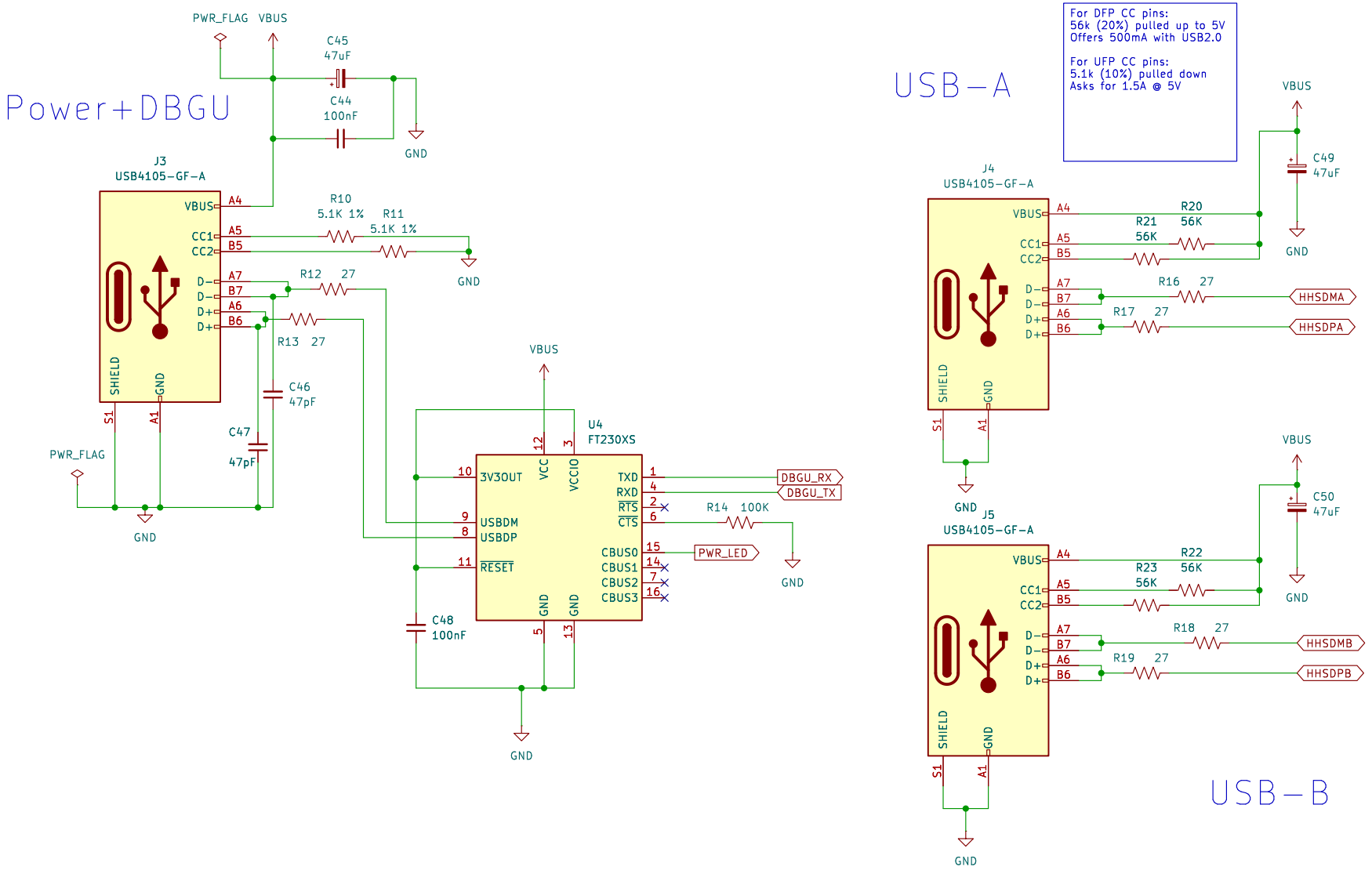


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KiCad E.D.A. 8.0.1	Rev: Id: 2/12

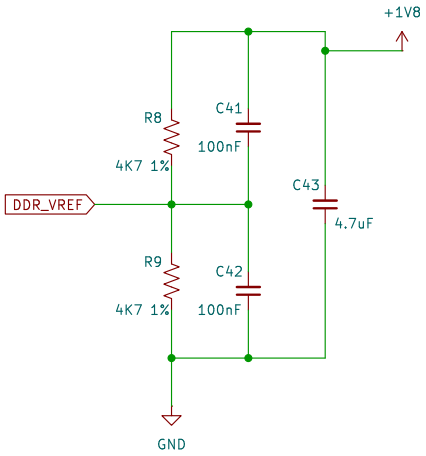


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Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 3/12


LCD2_EXT
J9
LCD2
J8
C79
1uF
GND
+3V3
LCD2_PWM
LCD2_WR
LCD2_RD
LCD2_CS
LCD2_A0
LCD2_RST
LCD2_DB0
LCD2_DB1
LCD2_DB2
LCD2_DB3
LCD2_DB4
LCD2_DB5
LCD2_DB6
LCD2_DB7
LCD2_WAIT
4.7uF
C77
GND
+3V3
L4
15uH
VBUS
D7
D_Schottky
1uF
C76
U8
SW
GND
FB
EN
VIN
MIC2287CYD5-TR
C70
1uF
J2 LCD1
R37
6.3
+3V3
C78
4.7uF
GND
GND
GND
U2C
SAM9X60D1G-I/4FB
PC31 U13 FIQ
PC30 T13 LCD1_PCK
PC29 P9 LCD1_DEN
PC28 P13 LCD1_HSYNC
PC27 U9 LCD1_VSYNC
PC26 R13 LCD1_PWM
PC25 P5
PC24 U14 LCD1_Disp
PC23 T9 LCD1_Blue7
PC22 R14 LCD1_Blue6
PC21 P8 LCD1_Blue5
PC20 T14 LCD1_Blue4
PC19 P6 LCD1_Blue3
PC18 R7 LCD1_Blue2
PC17 R8 LCD1_Blue1
PC16 R10 LCD1_Blue0
PC15 U5 LCD1_Green7
PC14 U2 LCD1_Green6
PC13 R4 LCD1_Green5
PC12 T5 LCD1_Green4
PC11 T3 LCD1_Green3
PC10 T8 LCD1_Green2
PC9 R1 LCD1_Green1
PC8 U7 LCD1_Green0
PC7 R2 LCD1_Red7
PC6 T7 LCD1_Red6
PC5 U4 LCD1_Red5
PC4 R5 LCD1_Red4
PC3 T1 LCD1_Red3
PC2 T4 LCD1_Red2
PC1 U3 LCD1_Red1
PC0 U8 LCD1_Red0
SAM9X60D1G LCD Pinout
PC31 FIQ
PC30 LCDPCK
PC29 LCDDEN
PC28 LCDHSYNC
PC27 LCDVSYNC
PC26 LCDPWM
PC25 –
PC24 LCDDISP
PC23 R[7]
PC22 R[6]
PC21 R[5]
PC20 R[4]
PC19 R[3]
PC18 R[2]
PC17 R[1]
PC16 R[0]
PC15 G[7]
PC14 G[6]
PC13 G[5]
PC12 G[4]
PC11 G[3]
PC10 G[2]
PC09 G[1]
PC08 G[0]
PC07 B[7]
PC06 B[6]
PC05 B[5]
PC04 B[4]
PC03 B[3]
PC02 B[2]
PC01 B[1]
PC00 B[0]
40pin LCD Pinout
1 LED Cathode–
2 LED Anode+
3 GND
4 VDD
5 R0
6 R1
7 R2
8 R3
9 R4
10 R5
11 R6
12 R7
13 G0
14 G1
15 G2
16 G3
17 G4
18 G5
19 G6
20 G7
21 B0
22 B1
23 B2
24 B3
25 B4
26 B5
27 B6
28 B7
29 GND
30 Clock
31 Disp On/Off
32 HSync (or GND)
33 VSync (or GND)
34 DE Data Enable
35 NC
36 GND
37 X1 Right
38 X2 Bottom
39 Y1 Left (SCL)
40 Y2 Top (SDA)
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.1
Id: 4/12



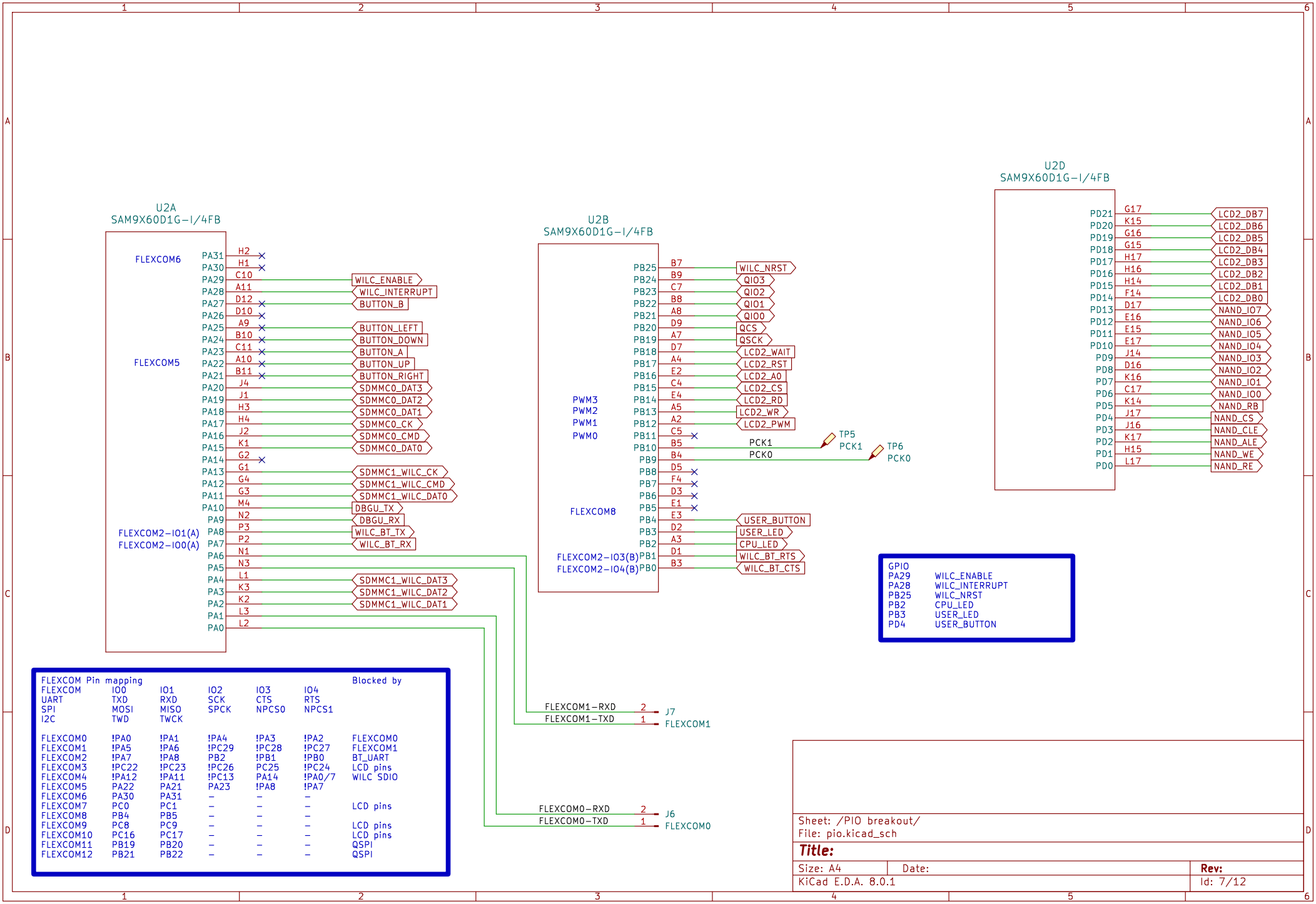
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KiCad E.D.A. 8.0.1		Id: 5/12

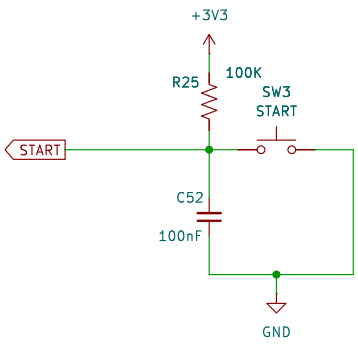


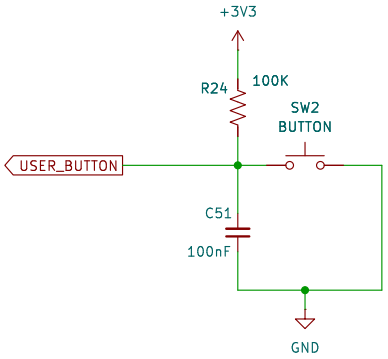
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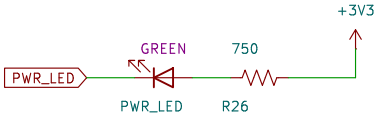
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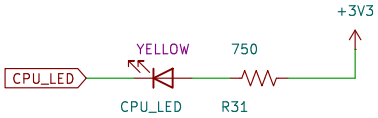
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KiCad E.D.A. 8.0.1		Id: 6/12

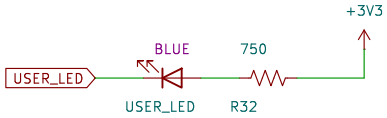

U2A
SAM9X60D1G-I/4FB
FLEXCOM6
FLEXCOM5
FLEXCOM2-IO1(A)
FLEXCOM2-IO0(A)
PA31 H2
PA30 H1
PA29 C10 WILC_ENABLE
PA28 A11 WILC_INTERRUPT
PA27 D12 BUTTON_B
PA26 D10
PA25 A9 BUTTON_LEFT
PA24 B10 BUTTON_DOWN
PA23 C11 BUTTON_A
PA22 A10 BUTTON_UP
PA21 B11 BUTTON_RIGHT
PA20 J4 SDMMC0_DAT3
PA19 J1 SDMMC0_DAT2
PA18 H3 SDMMC0_DAT1
PA17 H4 SDMMC0_CK
PA16 J2 SDMMC0_CMD
PA15 K1 SDMMC0_DAT0
PA14 G2
PA13 G1 SDMMC1_WILC_CK
PA12 G4 SDMMC1_WILC_CMD
PA11 G3 SDMMC1_WILC_DAT0
PA10 M4 DBGU_TX
PA9 N2 DBGU_RX
PA8 P3 WILC_BT_TX
PA7 P2 WILC_BT_RX
PA6 N1
PA5 N3
PA4 L1 SDMMC1_WILC_DAT3
PA3 K3 SDMMC1_WILC_DAT2
PA2 K2 SDMMC1_WILC_DAT1
PA1 L3
PA0 L2
U2B
SAM9X60D1G-I/4FB
PWM3
PWM2
PWM1
PWM0
FLEXCOM8
FLEXCOM2-IO3(B)
FLEXCOM2-IO4(B)
PB25 B7 WILC_NRST
PB24 B9 QIO3
PB23 C7 QIO2
PB22 B8 QIO1
PB21 A8 QIO0
PB20 D9 QCS
PB19 A7 QSCK
PB18 D7 LCD2_WAIT
PB17 A4 LCD2_RST
PB16 E2 LCD2_A0
PB15 C4 LCD2_CS
PB14 E4 LCD2_RD
PB13 A5 LCD2_WR
PB12 A2 LCD2_PWM
PB11 C5
PB10 B5 PCK1
PB9 B4 PCK0
PB8 D5
PB7 F4
PB6 D3
PB5 E1
PB4 E3 USER_BUTTON
PB3 D2 USER_LED
PB2 A3 CPU_LED
PB1 D1 WILC_BT_RTS
PB0 B3 WILC_BT_CTS
TP5
PCK1
TP6
PCK0
U2D
SAM9X60D1G-I/4FB
PD21 G17 LCD2_DB7
PD20 K15 LCD2_DB6
PD19 G16 LCD2_DB5
PD18 G15 LCD2_DB4
PD17 H17 LCD2_DB3
PD16 H16 LCD2_DB2
PD15 H14 LCD2_DB1
PD14 F14 LCD2_DB0
PD13 D17 NAND_IO7
PD12 E16 NAND_IO6
PD11 E15 NAND_IO5
PD10 E17 NAND_IO4
PD9 J14 NAND_IO3
PD8 D16 NAND_IO2
PD7 K16 NAND_IO1
PD6 C17 NAND_IO0
PD5 K14 NAND_RB
PD4 J17 NAND_CS
PD3 J16 NAND_CLE
PD2 K17 NAND_ALE
PD1 H15 NAND_WE
PD0 L17 NAND_RE
GPIO
PA29 WILC_ENABLE
PA28 WILC_INTERRUPT
PB25 WILC_NRST
PB2 CPU_LED
PB3 USER_LED
PD4 USER_BUTTON
FLEXCOM Pin mapping
FLEXCOM IO0 IO1 IO2 IO3 IO4 Blocked by
UART TXD RXD SCK CTS RTS
SPI MOSI MISO SPCK NPCS0 NPCS1
I2C TWD TWCK
FLEXCOM0 !PA0 !PA1 !PA4 !PA3 !PA2 FLEXCOM0
FLEXCOM1 !PA5 !PA6 !PC29 !PC28 !PC27 FLEXCOM1
FLEXCOM2 !PA7 !PA8 PB2 !PB1 !PB0 BT_UART
FLEXCOM3 !PC22 !PC23 !PC26 PC25 !PC24 LCD pins
FLEXCOM4 !PA12 !PA11 !PC13 PA14 !PA0/7 WILC SDIO
FLEXCOM5 PA22 PA21 PA23 !PA8 !PA7
FLEXCOM6 PA30 PA31 – – –
FLEXCOM7 PC0 PC1 – – – LCD pins
FLEXCOM8 PB4 PB5 – – –
FLEXCOM9 PC8 PC9 – – – LCD pins
FLEXCOM10 PC16 PC17 – – – LCD pins
FLEXCOM11 PB19 PB20 – – – QSPI
FLEXCOM12 PB21 PB22 – – – QSPI
FLEXCOM1-RXD 2
FLEXCOM1-TXD 1
J7
FLEXCOM1
FLEXCOM0-RXD 2
FLEXCOM0-TXD 1
J6
FLEXCOM0
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.1
Id: 7/12



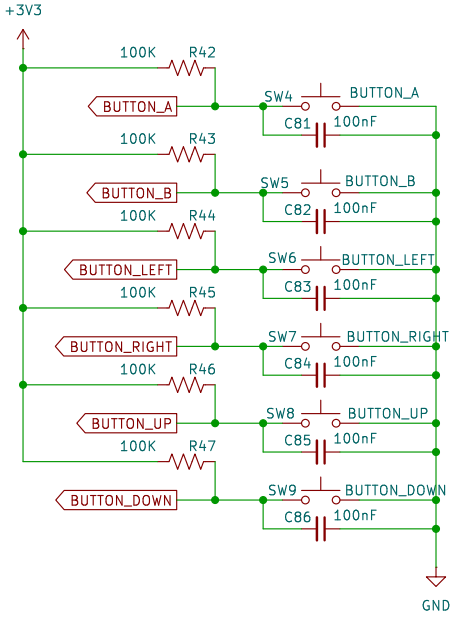








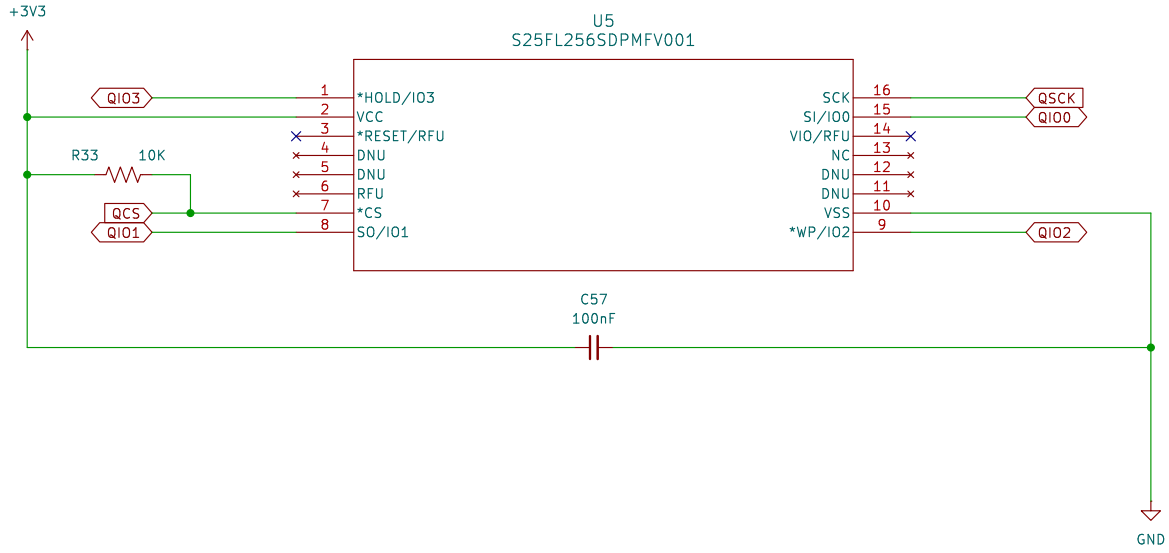
DPAD and Two buttons



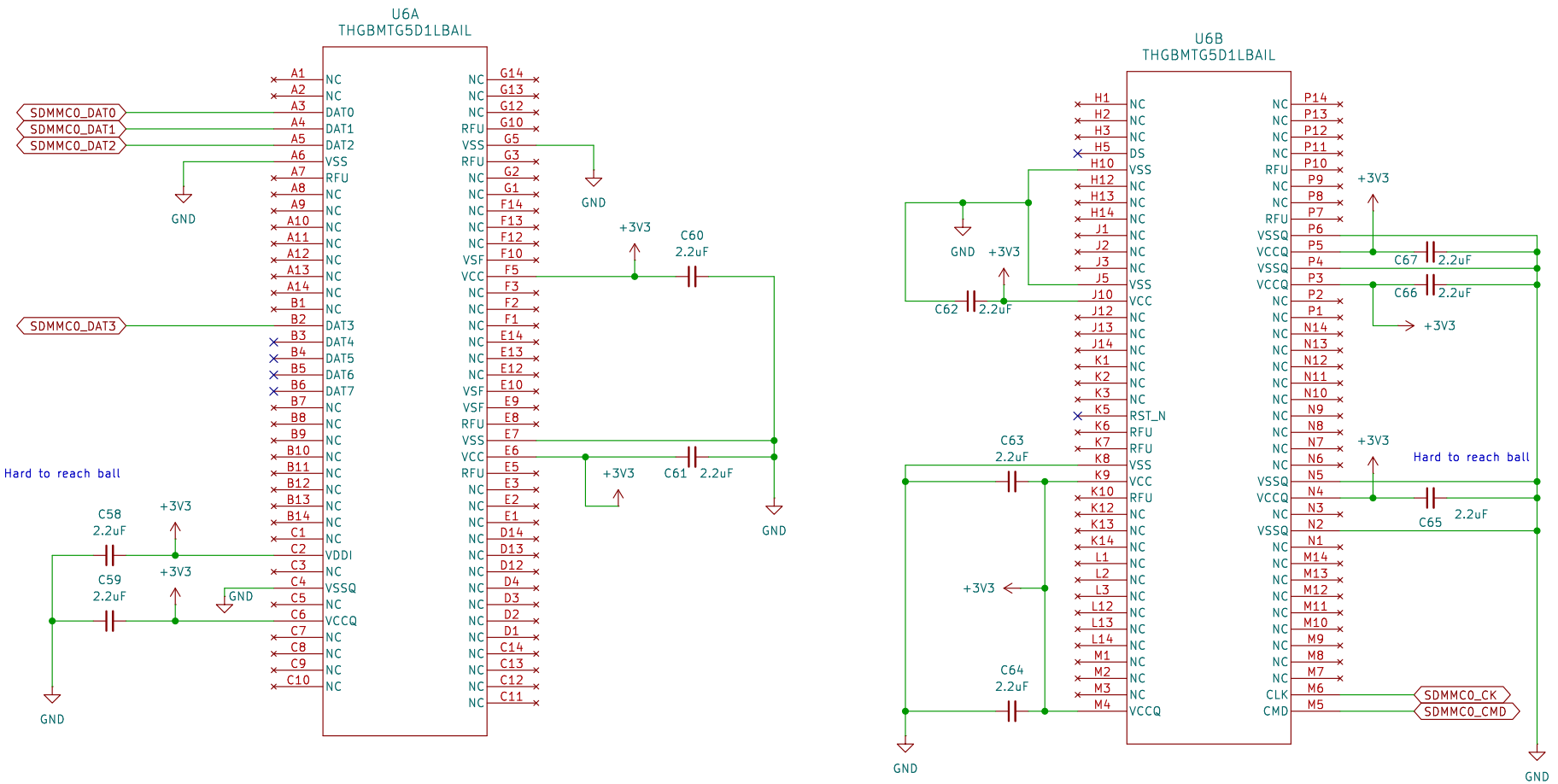
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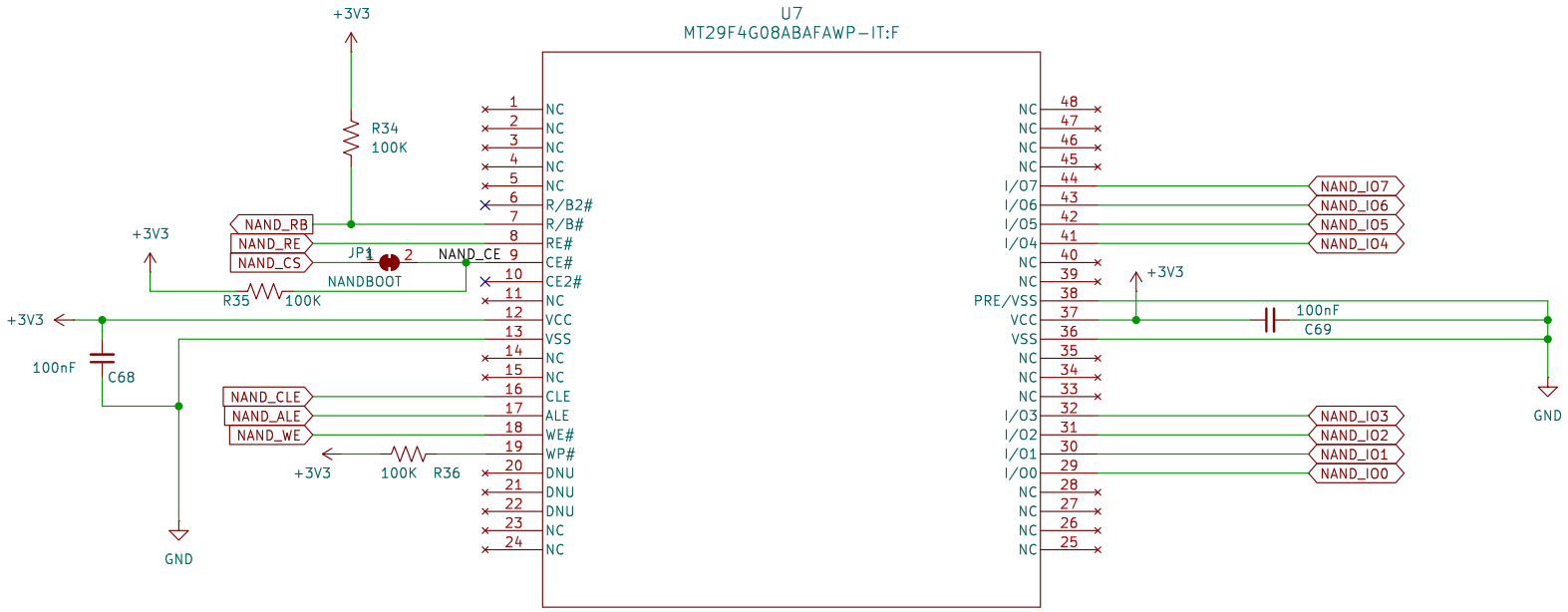
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 8/12



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Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 9/12



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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 10/12



Sheet: /NAND/		
File: nand.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.1		Id: 11/12

ATWILC3000
- SDIO mode
- VBAT and VDD are directly connected to +3V3
- BT UART connected with RTS/CTS Flexcom0
- WIFI debug UART only testpoints
- RTC not connected
- GPIO3/4 are LEDs
- Extra GPIO not connected

U9
ATWILC3000-MR110CA

1 GND
2 SDIO/SPI_CFG
3 NC
4 NC
5 NC
6 NC
7 RESETN
8 BT_TXD
9 BT_RXD
10 BT_RTS/I2C_SDA_S
11 BT_CTS/I2C_SCL_S
12 VDDIO
13 GND
14 GPIO3
15 GPIO4
16 UART_TXD
17 UART_RXD
18 VBAT

37 GND
36 GND
35 GPIO0
34 GPIO21
33 IRQN
32 GPIO20
31 GPIO19
30 GPIO18
29 GPIO17
28 GND
27 SD_DAT3/GPIO7
26 SD_DAT2/SPI_MOSI
25 SD_DAT1/SPI_SSN
24 SD_DAT0/SPI_MISO
23 SD_CMD/SPI_SCK
22 SD_CLK/GPIO8
21 GND
20 RTC_CLK
19 CHIP_EN

GND
GND
GND
WILC_INTERRUPT
GND
SDMMC1_WILC_DAT3
SDMMC1_WILC_DAT2
SDMMC1_WILC_DAT1
SDMMC1_WILC_DAT0
SDMMC1_WILC_CMD
SDMMC1_WILC_CK
GND
WILC_ENABLE

WILC_NRST
WILC_BT_TX
WILC_BT_RX
WILC_BT_CTS
WILC_BT_RTS
WILC_LED_GREEN
WILC_LED_RED
TP3 WILC_WIFI_TXD
TP4 WILC_WIFI_RXD

+3V3
C71 100nF
C72 100nF
C73 47uF
C74 2.2uF
GND

R38
100K

+3V3
Y3
4 VDD
1 OUTPUT
OE 3
GND 2
ASH7KW-32.768KHZ-L-T
C75
100nF
GND

+3V3
R39
1K
GREEN
WILC_LED_GREEN
WILC_LED_GREEN
R40
1K
RED
WILC_LED_RED
WILC_LED_RED

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Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.1
Id: 12/12